

Two Elementary Mean-Class Constructions

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Status: candidate full resolution of Open Question 10 and of the literal simple-existence clause of Open Question 3 in arXiv:1403.7430.

1 Questions and notation

For a class A of functions on J , Guenzler defines

$$M_h f(t) = \frac{1}{h} \int_0^h f(t+s) ds, \quad MA = \{f \in L^1_{\text{loc}}(J) : M_h f \in A \text{ for every } h > 0\}.$$

The class A has property (Δ) if

$$\Delta_a f = f(\cdot + a) - f \in A \quad (a > 0) \implies f - M_h f \in A \quad (h > 0).$$

Open Question 3 asks for a simple $A \subset MA$ without (Δ) , preferably linear, invariant, uniformly closed and contained in BC . Open Question 10 asks for U, V such that

$$MU + MV \subsetneq M(U + V).$$

Both questions occur on PDF page 30 of the source [1].

1. Does there for $J = \mathbb{R}_+$ exist a non-trivial λ -class satisfying the Loomis condition (L_b) but $\neq BUC(J, X)$?
(λ -class: linear closed subspace of BUC with (Γ) (see § 8) and in a suitable sense invariant [4] p. 117].)
2. Does there exist a simpler proof for (Δ) for $BUC + \gamma BUC$ than ours in §6?
3. Does there exist a simple example of an A with $A \subset MA$ but without (Δ) , and if possible linear, invariant, uniformly closed, $\subset BC$?
(See example 4.8, and Q4.)
4. Is there an extension of Proposition 4.1 which explains why reasonable $A \subset MA$ and $\subset BC$ but without (Δ) are rare?
5. Is there a simple proof of (Δ) for $gAAP$? Extension to $g_{\omega,r}AAP, \gamma AAP$?
(True by an unpublished result of B. Basit and the author 2002.)
6. Is there a simple example of a (linear, invariant, uniformly closed) $A \neq \{0\}$ with $A \subset MA$ and (P_b) , but without (L_b) ?
(Definitions: (6.4), (6.5); the converse situation holds e.g. for $X_c + gX$, or $AP + gX$; see Q7.)
7. Is there a theorem which explains why reasonable A with (P_b) but not (L_b) are rare? (see Q6., Q4.)
8. Does $UC + gUC$ satisfy (Δ) ?
9. Which of the other two “?” in the Δ -matrix in § 6 can be replaced by a “+”?
($UC + gAP = UC + gAAP$, Q8.)
10. Examples of $U, V \subset X^J$ with $MU + MV \subsetneq M(U + V)$?
(See [6] Prop.5.1 (ii) p. 38]?)
11. Can one improve Example 3.18 to $g_{\omega,r}UC \subset M(g_{\omega,r}U)$ with U smaller than BUC ?

As in the source, mean classes cannot distinguish functions equal almost everywhere. Write $\mathcal{N}$ for the class of locally integrable functions that vanish almost everywhere.

2 Proof intuition

For Question 3, the simplest mean-stable object is an order cone: averaging preserves nonnegativity. A strictly increasing bounded function has every positive difference in that cone, whereas subtracting a forward average produces a strictly negative function. This answers the existence request with an invariant uniformly closed subset of BC ; only the optional linearity refinement remains.

For Question 10, translations rotate the plane spanned by $\cos t$ and $\sin t$. Consequently averaging preserves that plane, although it does not preserve either coordinate axis. Requiring *every* forward mean to lie on one axis is so rigid that the original function must vanish almost everywhere. Thus the two individual mean classes are null while the mean class of their sum contains the whole trigonometric plane.

3 The two constructions

Theorem 1. *Let $J = \mathbb{R}$ and use real scalar-valued functions.*

1. Define

$$A = \{g \in BC(\mathbb{R}) : g(t) \geq 0 \text{ for all } t\}.$$

Then A is invariant, uniformly closed, contained in $BC(\mathbb{R})$, satisfies $A \subset MA$, and does not satisfy (Δ) .

2. Let

$$U = \text{span}\{\cos t\}, \quad V = \text{span}\{\sin t\}.$$

Then

$$MU = MV = \mathcal{N}, \quad U + V \subset M(U + V),$$

and hence $MU + MV \subsetneq M(U + V)$.

Proof. For the first assertion, translation and uniform limits preserve nonnegativity. If $g \in A$, then $M_h g$ is bounded, continuous and nonnegative for every $h > 0$, so $g \in MA$.

Now take $f(t) = \tanh t$. Since f is strictly increasing,

$$\Delta_a f(t) = \tanh(t + a) - \tanh t > 0 \quad (t \in \mathbb{R}, a > 0),$$

so every positive difference belongs to A . On the other hand,

$$M_h f(t) > f(t) \quad (t \in \mathbb{R}, h > 0),$$

because the integrand $f(t + s)$ is strictly larger than $f(t)$ for $0 < s \leq h$. Therefore $f - M_h f$ is strictly negative and is not in A . Thus (Δ) fails.

For the second assertion, direct integration gives

$$M_h(\cos t) = \frac{\sin h}{h} \cos t + \frac{\cos h - 1}{h} \sin t, \tag{1}$$

$$M_h(\sin t) = \frac{1 - \cos h}{h} \cos t + \frac{\sin h}{h} \sin t. \tag{2}$$

Thus every mean of every element of $U + V$ again belongs to $U + V$, proving $U + V \subset M(U + V)$.

It remains to compute the individual mean classes. Suppose $g \in MU$. For every $h > 0$ there is a scalar c_h such that

$$M_h g(t) = c_h \cos t.$$

Choose a Lebesgue point t_0 of g with $\cos t_0 \neq 0$. As $h \downarrow 0$, the one-sided differentiation theorem gives

$$c_h = \frac{M_h g(t_0)}{\cos t_0} \longrightarrow c := \frac{g(t_0)}{\cos t_0}.$$

At every Lebesgue point t of g , it follows that $g(t) = \lim_{h \downarrow 0} c_h \cos t = c \cos t$. Hence $g = c \cos t$ almost everywhere. Formula (1) with $h = \pi$ can lie in U only when its sine coefficient $-2c/\pi$ vanishes. Thus $c = 0$, proving $MU = \mathcal{N}$.

The same argument for $g \in MV$ first gives $g = c \sin t$ almost everywhere. Formula (2) at $h = \pi$ has cosine coefficient $2c/\pi$, so $c = 0$ and $MV = \mathcal{N}$. Consequently $MU + MV = \mathcal{N}$, whereas $\cos t \in U + V \subset M(U + V)$ and $\cos t \notin \mathcal{N}$. The inclusion is strict. $\square$

Checks, novelty, and scope

The proof uses only elementary integration and one-sided Lebesgue differentiation. The null class is retained explicitly because the source's definition of MA uses locally integrable representatives and likewise records null functions in several examples. If spaces are taken modulo almost-everywhere equality, $\mathcal{N}$ is simply the zero element.

The Question 3 example satisfies every requested adjective except linearity: it is an invariant uniformly closed cone in $BC(\mathbb{R})$. Hence it fully answers the literal request for a simple example but does not improve the source's more elaborate linear examples to an equally elementary linear one. The Question 10 construction answers the displayed strict-inclusion problem with two one-dimensional spaces; replacing t by any nonzero frequency λt gives the same family of examples.

A bounded novelty search checked the run indexes, the exact source title and arXiv identifier, the exact Question 10 inclusion, and combinations of “mean class”, $\text{span}\{\cos, \sin\}$ and the source author's name. No later explicit answer or duplicate was located. Citation-database coverage of this arXiv-only note was weak, so novelty remains provisional.

References

- [1] H. Guenzler, *Spaces of rapidly oscillating functions*, arXiv:1403.7430v1 (2014), especially Section 9, Open Questions 3 and 10, PDF p. 30.